

1 The iodine clock

EXPERIMENT 1

Characterization of the amylase enzyme

Goal

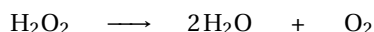
The goal of this experiment is to use the initial rate method to obtain the Michaelis-Menten constants for the starch hydrolysis catalyzed by amylase.

Background

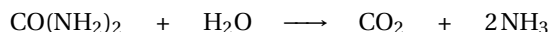
Enzymes are biological catalysts responsible for the speeding up of biochemical reactions. *Catalase* or *amylase* are two enzymes examples; the former is responsible for breaking down hydrogen peroxide (H_2O_2) whereas the latter catalyzes the hydrolysis of starch. Enzymes are particularly relevant in biochemistry accelerating chemical reactions at rates necessary for life. Most biochemical reactions require an enzyme. This section covers the chemical kinetic principles behind biological catalysis and the factors that impact enzyme activity.

What are enzymes

Enzymes are large biological molecules, in general proteins, able to speed up chemical reactions. Enzymes are more efficient than regular catalysts. The activation energy of enzyme-catalyzed chemical reactions tends to be smaller than that of regularly-catalyzed reactions. For example, the activation energy of the Fe^{2+} -catalyzed decomposition of hydrogen peroxide



is 42 KJ/mol, whereas that for the catalase-catalyzed decomposition is only 7 KJ/mol. As enzymes are large molecules, only a small part of the molecule called the active site is actively involved in the catalytic process. Enzymes are associated with coenzymes which are inorganic compounds essential for catalyst action. On one hand, we can find very specific catalysts. An example would be urease which only catalyzes the hydrolysis of urea ($\text{CO}(\text{NH}_2)_2$), the reaction between urea and water:



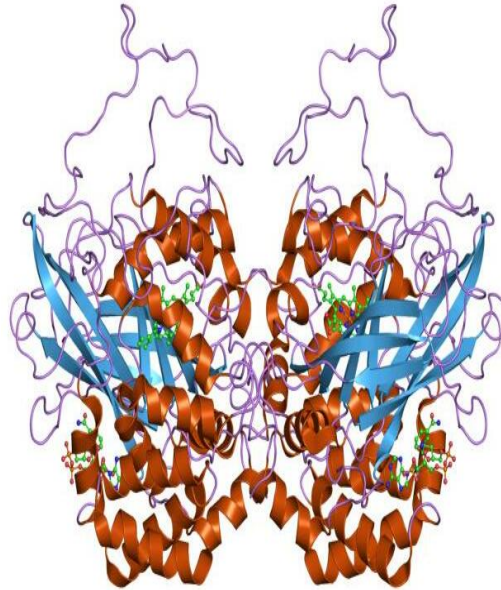
On the other hand, we can also find less specific enzymes such as proteolytic enzymes which catalyze the hydrolysis of the peptide linkage that connects amino acids in proteins. In fact, these types of enzymes are group-specific and will catalyze the hydrolysis of the peptide linkage in any molecule with such a chemical bond.

▼Amylase



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▼Catalase



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Amylase

Amylase is a digestive enzyme involved in the hydrolysis of complex carbohydrates (like starch) into simpler sugars. It is primarily produced by the pancreas and salivary glands aiding digestion. It is present in human saliva, and it was named after the Ancient Greek name for saliva. It exists in three different forms: α -amylase, β -amylase, and γ -amylase. α -amylase is the major form of amylase found in humans and other mammals, also being the fastest. β -amylase is present in plants and microbes, being responsible for the ripening of fruit as it breaks starch into maltose, resulting in the sweet flavor of ripe fruit. γ -amylase, present in animals and microbes, is most active around pH 3, being the most acidic optimum pH of all amylases.

Michaelis-Menten equation

The Michaelis-Menten equation gives the rate of an enzyme-catalyzed reaction in terms of the concentration of the enzyme.

$$v = \frac{V_m [S]}{K_m + [S]} \quad (1)$$

where:

v is the enzyme-catalyzed reaction rate

$[S]$ is the enzyme concentration

V_m is the maximal velocity

K_m Michaelis constant

It involves two different constants V_m which is the maximal rate and K_m , the Michaelis constant. Enzymes speed up chemical rate. However, when the enzyme concentration is too large, the enzyme catalyzed rate reaches a plateau with a maximal rate. At this point increasing the enzyme concentration does not increase the reaction rate. K_m is a constant that involves the reaction rate constant of the enzyme-catalyzed reaction mechanism.

Lineweaver-Burk linear equation

The Michaelis-Menten equation is not a linear relationship and hence it is not suitable to fit experimental data. Linear relations are suitable for obtaining kinetic data because they are simple to analyze when dealing with large datasets. Perhaps

more importantly, errors are well-understood in linear expressions. The quality of the linear fit simply relates to the correlation coefficient.

Hence, the Michaelis-Menten equation can be recast into a first linear expression by simply using reciprocals. The resulting equation in which the inverse of the rate $1/r$ (vertical axis) is plotted against the inverse of $1/[S]$ (horizontal axis) is called the Lineweaver-Burk equation

$$\frac{1}{v} = \frac{K_m}{V} \cdot \frac{1}{[S]} + \frac{1}{V} \quad (2)$$

where:

$1/v$ is the inverse of the enzyme-catalyzed reaction rate

$1/[S]$ is the inverse of the enzyme concentration

V_m is the maximal velocity

K_m Michaelis constant

The slope of this plot is negative. From the intercept we can obtain V_m and from the slope we can calculate K_m .

Eadie linear equation

The Michaelis-Menten equation can be recast into a second linear expression as shown below:

$$v = V_m - K_m \cdot \frac{v}{[S]} \quad (3)$$

is called the Eadie equation. In the Eadie equation, the rate $1/r$ (vertical axis) is plotted against $v/[S]$ (horizontal axis). The slope of this plot is negative. The advantage of this equation is that it spreads more the data than the Lineweaver-Burk equation.

PH and temperature effects

Enzyme activity depends on pH and temperature. pH has a strong impact on enzyme activity passing through a maximum as pH increases. Similarly, enzyme activity also passes through a maximum as temperature increases. Enzymes show maximum activity at an *optimum pH* and *optimum temperature*. This maximum depends on the nature of the enzyme and the catalyzed reaction. They reversibly lose activity at pH and temperature slightly lower or higher than their optimum. Enzyme deactivation is irreversible if the acidity or basicity is too high or if temperature is too high or low. One explanation for the pH effects is that ionization of protein carboxylic and amino groups occurs when pH is not taken too far within the optimum pH. Differently, enzymes lose their tertiary structure hence losing their activity at very low or high pH values or temperature higher than 30°C.

Because of their shape, graphs displaying catalyst activity vs pH or temperature are often referred to as volcano plots. Figure ?? displays pH and temperature volcano plots for the activity of amylase.

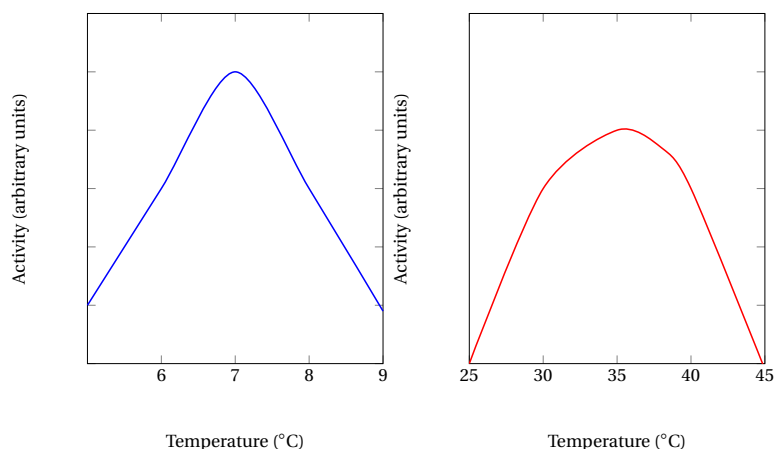


Figure ?? Schematic representation of pH (left) and temperature (right) volcano plots for the activity of amylase. Amylase is the digestive enzyme that breaks down complex carbohydrates (like starch) into simpler sugars, aiding digestion. The optimum temperature is 37°C, and the optimum pH range is 6.7-7.0.

The experiment

This experiment is divided into the following parts: obtaining the reaction rate constant, evaluating the influence of the temperature on the reaction rate, and evaluating the influence of a catalyst on the reaction rate.

Part A. Obtaining the reaction rate constant.

The reaction to be studied and its rate law are:

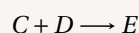
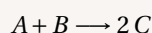


Using the **initial rates method** the reactants orders, a and b , can be solved. Find the suitable combinations of experiments where only one initial concentration of $\text{S}_2\text{O}_8^{2-}$ or I^- changes and compare the rates using a fraction. The initial concentrations must be recalculated using the volumes and their original concentrations. Once the reactant orders are known and the initial concentrations of $\text{S}_2\text{O}_8^{2-}$ and I^- for each reaction have been recalculated, the reaction constant, k , can be solved. See the example for more details:

	Volume					
	NaI	NaCl	Na ₂ S ₂ O ₃	Starch	K ₂ SO ₄	K ₂ S ₂ O ₈
	0.2M	0.2M	0.01M	2%	0.2M	0.2M
Exp. 1	2.0	2.0	2.0	1.0	2.0	2.0
Exp. 2	2.0	2.0	2.0	1.0	0	4.0
Exp. 3	4.0	0	2.0	1.0	2.0	2.0

Example

The reaction starts from a solution containing A (1 M) and B (0.5 M) and D (1 M). There are two reactions occurring in the solution. A slow reaction where C is produced and a fast reaction where C is reduced to E ions by D:



An indicator is added, such that the solution will turn dark in the presence of substance C.

Three experiments are performed using different initial volumes and the following times for the solution to turn dark are measured;

	Volumes (mL)				time
	A	B	D	Water	(s)
Exp. 1	4.0	2.0	2.0	2.0	20
Exp. 2	4.0	1.0	2.0	2.0	40
Exp. 3	2.0	1.0	2.0	4.0	160

Calculate the reactant orders and the reaction rate constant.

Answer: First write the reaction rate expression: $\text{Rate} = k[A]^a[B]^b$

Recalculate the concentrations for each reactant. Use the formula for dilutions $M_1 V_1 = M_2 V_2$, where V_2 is the total volume $V_A + V_B + V_D + V_{\text{water}}$.

For example, to calculate the concentration of A in Experiment 1:

$$M_1 V_1 = M_2 V_2 \quad M_2 = \frac{M_1 V_1}{V_2} \quad M_2 = \frac{1M \cdot 4mL}{10mL} = 0.4M$$

	[A]	[B]	[D]	rate (Ms ⁻¹)
Exp. 1	0.4 M	0.2 M	0.2 M	$5 \cdot 10^{-3}$
Exp. 2	0.4 M	0.1 M	0.2 M	$2.5 \cdot 10^{-3}$
Exp. 3	0.2 M	0.1 M	0.2 M	$6.25 \cdot 10^{-4}$

The reaction rate is the change in concentration per unit time. The change in concentration is determined by the second reaction (the clock). For the first experiment it takes 20 seconds to turn black. In those 20 seconds all of D has reacted, i.e. 0.2 M in 20 seconds.

Following the stoichiometry of the reaction, if 1 mole of D reacts with 1 mole of C and every 2 moles of C are produced when 1 mole of A reacts. The relation between D and A is 2:1.

$$1 \text{ mole of A} = 1 \text{ mole A} \cdot \frac{2 \text{ moles C}}{1 \text{ moles A}} \cdot \frac{1 \text{ mole D}}{1 \text{ mole C}} = 2 \text{ moles of D}$$

Therefore, 0.2 M of D corresponds to 0.1 M of A reacting in 20 s. The rate for experiment 1 is $5 \cdot 10^{-3}$ M/s. The order of the reactants are calculated by comparing the rate laws. Each time, choose two experiments where only one parameter is different.

$$\frac{\text{Rate 2}}{\text{Rate 3}} = \frac{k[A]^a[B]^b}{k[A]^a[B]^b} = \frac{k 0.4^a 0.1^b}{k 0.2^a 0.1^b} = \frac{2.5 \cdot 10^{-3}}{6.25 \cdot 10^{-4}}$$

$$2^a = 4 \quad a = 2$$

Similarly, using rate 2 over rate 1, $b=1$ is obtained.

The reactant order can be used to obtain the reaction rate.

$$\text{Rate} = k[A]^a[B]^b \quad k = \frac{\text{Rate}}{[A]^a[B]^b} = \frac{5 \cdot 10^{-3} \text{ M s}^{-1}}{(0.4 \text{ M})^2 (0.1 \text{ M})^1} = 0.15625 \text{ M}^{-2} \text{ s}^{-1}$$

Since experimentally it is more likely to obtain one slightly different k for each experiment, an average should be calculated.

Parts B. Evaluating the influence of the temperature on the reaction rate.

According to **Arrhenius equation**, the rate constant is affected by the temperature following the equation:

$$k = Ae^{-E_a/RT}$$

where A is a constant, E_a is the activation energy, R is the universal gas constant, and T is the temperature. A higher temperature would make the negative exponent smaller, and therefore the k would be greater. In other words, an increase in the temperature should accelerate the reaction. It is possible to calculate the activation energy by measuring the reaction rates at two different temperatures. After calculating the respective rate constants these are compared in their logarithmic forms:

$$\ln(k) = -\frac{E_a}{RT} + \ln(A)$$

After subtracting the expression above for both temperatures, T_1 and T_2 , the following equation is generated:

$$\ln\left(\frac{k_2}{k_1}\right) = \frac{E_a}{R} \left(\frac{1}{T_1} - \frac{1}{T_2}\right)$$

The values of k_1 and k_2 obtained respectively at T_1 and T_2 are used to calculate E_a . The temperatures should be given in Kelvins since $R = 8.3145 \text{ J/K mol}$.

Part C. Evaluating the influence of a catalyst on the reaction rate.

A catalyst is a substance that without being consumed, accelerates the reaction. The catalyst generates a new reaction path with a lower activation energy. Looking at the Arrhenius equation, the effect of lowering the activation energy works in the same way as an increase in temperature. The catalyst used in this experiment will be copper sulfate (CuSO_4). The Cu(II) cations are responsible for the catalytic effect.

Procedure

Part A. Obtaining the rate law. Preparing the mixtures

- ☐ *Step 1:* – Obtain three large test tubes with their corresponding stoppers, a suitable grid or rack, a Mohr pipet, and a thermometer.
- ☐ *Step 2:* – Label the test tubes with numbers 1, 2 and 3.
- ☐ *Step 3:* – Read the Good Lab Practice box

Good Lab Practice

- ✍ Pipettes are always used together with a suction bulb or a syringe. **Never** such a chemical using the mouth!
- ✍ **Never** leave a pipette laying on the table **while connected to the syringe**. Remaining liquid can leak into the syringe. The syringe will be damaged and the liquid will return contaminated to the pipette.
- ✍ Always wash pipettes with distilled water and rinse them using the same product to be measured. Such a small volume, disconnect the syringe and move the liquid about displacing the distilled water. Discard the rinsing liquid as waste. Rinse twice.

- ☐ *Step 4:* – Obtain about 25 mL **0.20 M NaI** in a small beaker.
- ☐ *Step 5:* – Clean and rinse the pipette with NaI. Pipette 2.0 mL in test tube 1, 2.0 mL in test tube 2 and 4.0 mL in test tube 3. Refer to the table below (Table of quantities) for the reagent quantities to add.
- ☐ *Step 6:* – Obtain no more than 20 mL **0.20 M NaCl** in a small beaker.
- ☐ *Step 7:* – Clean and rinse the pipette with NaCl. Pipette 2.0 mL in test tube 1 and 2.0 mL in test tube 2. Nothing in test tube 3. Refer to the table below (Table of quantities) for the reagent quantities to add.
- ☐ *Step 8:* – Obtain no more than 20 mL **0.010 M Na₂S₂O₃** in a small beaker.
- ☐ *Step 9:* – Clean and rinse the pipette with Na₂S₂O₃. Pipette 2.0 mL in each test tube. Refer to the table below (Table of quantities) for the reagent quantities to add.
- ☐ *Step 10:* – Obtain approximately 10 mL **2% starch solution** in a small beaker.
- ☐ *Step 11:* – Clean and rinse the pipette with starch. Pipette 1.0 mL in each test tube. Refer to the table below (Table of quantities) for the reagent quantities to add.
- ☐ *Step 12:* – Obtain no more than 20 mL **0.20 M K₂SO₄** in a small beaker.
- ☐ *Step 13:* – Clean and rinse the pipette with K₂SO₄. Pipette 2.0 mL in test tube 1, nothing in test tube 2 and 2.0 mL in test tube 3. Refer to the table below (Table of quantities) for the reagent quantities to add.
- ☐ *Step 14:* – Check that the volumes in the 3 test tubes make sense with the following table. Test tubes 1 and 3 should have the same volume. Test tube 2 should have slightly less volume.
- ☐ *Step 15:* – Using the thermometer, record the room temperature in Kelvins.

Table of quantities						
	Volume (mL)					
	NaI	NaCl	Na ₂ S ₂ O ₃	Starch	K ₂ SO ₄	K ₂ S ₂ O ₈
	0.2M	0.2M	0.01M	2%	0.2M	0.2M
Exp. 1	2.0	2.0	2.0	1.0	2.0	2.0
Exp. 2	2.0	2.0	2.0	1.0	0	4.0
Exp. 3	4.0	0	2.0	1.0	2.0	2.0

Part A. Obtaining the rate law. Mixing the test tubes

- ☐ *Step 1:* – Now you are about to start up the reaction. It is critical to follow the steps below.
- ☐ *Step 2:* – Every reaction should be done in **the same way** to be comparable. The same steps, in the same order and at the same pace should be followed.
- ☐ *Step 3:* – You will be adding $K_2S_2O_8$. After adding the $K_2S_2O_8$, the test tube should be closed with the stopper, and the content should be mixed by inverting the test tube 16 times. Do not rush, turn the test tube *Mississippily*. Do the same for the following reactions.

CAUTION!

-  The stopper should always be placed before inverting the test tube.
-  Failure to place the stopper will cause the spill of the reaction.
-  There is a story behind every Caution Box.

Part A. Obtaining the rate law. Starting the reaction

- ☐ *Step 1:* – Obtain about 25 mL **0.20 M $K_2S_2O_8$** in a small beaker. Make sure you do not contaminate the bottle of reagent by always retrying out of the bottle into a beaker and never pipetting directly from the bottle. Clean and rinse the pipette with $K_2S_2O_8$.
- ☐ *Step 2:* – Get test tube 1 and a stopper. Set the time to zero.
- ☐ *Step 3:* – Start the stopwatch as you pipette 2.0 mL of $K_2S_2O_8$ in test tube 1. Time should start to be measured as soon as you add the first drop of $K_2S_2O_8$ in a test tube. Mix the reaction as explained above.
- ☐ *Step 4:* – Stop the stopwatch when the solution turns dark. The color will appear suddenly. If more than 5 minutes elapse without the solution turning dark, start doubting about what have you mixed.
- ☐ *Step 5:* – Repeat the previous steps now for test tube 2 and pipetting this time 4.0 mL of $K_2S_2O_8$ into the test tube. .
- ☐ *Step 6:* – Repeat one more time the previous steps now for test tube 3 and pipetting this time 2.0 mL of $K_2S_2O_8$ into the test tube.
- ☐ *Step 7:* – Now you need to repeat all steps above (mixture preparation and reaction) to do a second trial for the three test tubes.
- ☐ *Step 8:* – Discard the solutions in the indicated waste container and clean the test tubes.

Part B. Evaluating the influence of the temperature on the reaction rate.

- ☐ *Step 1:* – Label three large test tubes with the letters “T”, “C” and “K”. “T” stands for “Temperature”, “C” for “Catalyst” and “K” for “ $K_2S_2O_8$ ”.
- ☐ *Step 2:* – Add to test tubes “T” and “C” the same volumes of NaI, NaCl, $Na_2S_2O_3$, Starch and K_2SO_4 as for test tube 1. Do not add the $K_2S_2O_8$ yet. Reserve test tube “C” in the rack for Part C.
- ☐ *Step 3:* – Add about 5 mL of $K_2S_2O_8$ to test tube “K”.
- ☐ *Step 4:* – Obtain a 600 mL beaker and, with the help of the thermometer, add hot and cold water until you have about 400 mL of water between 40 and 45 degrees Celsius.

- ☐ *Step 5:* – Place the test tubes “T” and “K” in the hot water bath and allow them to temper for at least 5 minutes. Part C can be done during those 5 minutes. .
- ☐ *Step 6:* – Measure the temperature of the water after the tempering time. Write down this temperature in the results table. Start the reaction by pipetting 2 mL of $K_2S_2O_8$ from test tube “K” into test tube “T”. After the 16-Mississippi mixing, return the test tube to the bath and wait to record the time it takes for the solution to turn dark.

Part C. Evaluating the influence of a catalyst on the reaction rate.

- ☐ *Step 1:* – Add **one drop** of **0.2 M $CuSO_4$** to test tube “C”. Shake the test tube to mix the solution.
- ☐ *Step 2:* – Start the stopwatch as you pipette 2.0 mL of $K_2S_2O_8$ in the test tube. Mix the reaction as explained above.
- ☐ *Step 3:* – Record the time it takes for the solution to turn dark.

Calculations

- ① This is lab temperature. You will use this value for Part B as well.
- ① This is the volume of iodine added in each experiment.
- ② This is the volume of thiosulfate added in each experiment.
- ③ This is the volume of peroxydisulfate added in each experiment.
- ④ This is the total volume v_{total} of the reacting mixture.
- ⑤ This is the peroxydisulfate concentration in the mixture:

$$[S_2O_8^{-2}] = \frac{v_{S_2O_8^{-2}} \cdot c_{S_2O_8^{-2}}}{v_{total}} = \frac{\textcircled{3} \cdot 0.2}{\textcircled{4}}$$

- ⑥ This is the iodine concentration in the mixture:

$$[I^-] = \frac{v_{I^-} \cdot c_{I^-}}{v_{total}} = \frac{\textcircled{1} \cdot 0.2}{\textcircled{4}}$$

- ⑦ This is the change in thiosulfate concentration during the reaction:

$$\Delta[S_2O_3^{-2}] = \frac{v_{S_2O_3^{-2}} \cdot c_{S_2O_3^{-2}}}{v_{total}} = \frac{\textcircled{2} \cdot 0.01}{\textcircled{4}}$$

- ⑧ This is the average time Δt for each experiment.

- ⑨ This is the reaction rate:

$$r = \frac{\Delta[S_2O_3^{-2}]}{2 \cdot \Delta t} = \frac{\textcircled{7}}{2 \cdot \textcircled{8}}$$

- ⑩ This is the peroxydisulfate order in the rate law:

$$a = \frac{\ln\left(\frac{r_{Exp1}}{r_{Exp2}}\right)}{\ln\left(\frac{[S_2O_8^{-2}]_{Exp1}}{[S_2O_8^{-2}]_{Exp2}}\right)} = \frac{\ln\left(\frac{(9)_{Exp1}}{(9)_{Exp2}}\right)}{\ln\left(\frac{(5)_{Exp1}}{(5)_{Exp2}}\right)}$$

- (11) This is the iodide order in the rate law:

$$b = \frac{\ln\left(\frac{r_{Exp1}}{r_{Exp3}}\right)}{\ln\left(\frac{[I^-]_{Exp1}}{[I^-]_{Exp3}}\right)} = \frac{\ln\left(\frac{(9)_{Exp1}}{(9)_{Exp3}}\right)}{\ln\left(\frac{(6)_{Exp1}}{(6)_{Exp3}}\right)}$$

- (12) This is the rate constant calculated for each experiment:

$$k = \frac{r}{[S_2O_8^{-2}] \cdot [I^-]} = \frac{(9)}{(5) \cdot (6)}$$

- (13) This is the average rate constant.

- (14) The data on this row should be the same as in the first row in the Results table of Part B.

- (15) This is the activation energy. You need to use the two different reaction constants at different temperatures. R is 3.184 J/K·mol:

$$E_a = \frac{T_1 \cdot T_2}{T_2 - T_1} \cdot R \cdot \ln\left(\frac{k_2}{k_1}\right)$$

STUDENT INFO

Name:

Date:

Pre-lab Questions

Characterization of the amylase enzyme

1. What is the difference between a test tube and a jumbo test tube?
2. For the reaction $2A + 3B \rightarrow C$ what is the mathematical expression that relates the rate of A (r_a) and the rate of B (r_b)?
3. Explain the Iodine clock reaction used in this experiment.
4. What is starch and where can you find it in nature?
5. Low-quality boiled ham often contains potatoes to make it cheaper. How could you test if one piece of ham is pure meat or it contains potatoes?
6. Find out the difference between a **Mohr pipette** and a **transfer pipette**.

Name:

Date:

Results

EXPERIMENT 1

Characterization of
the amylase enzyme

0 Laboratory Temperature _____K

Trial	Exp.	C_{NaI}	C_{NaCl}	$C_{\text{Na}_2\text{S}_2\text{O}_3}$	C_{Starch}	$C_{\text{K}_2\text{SO}_4}$	$C_{\text{K}_2\text{S}_2\text{O}_8}$	v_{Total}	time	$[\text{S}_2\text{O}_8^{-2}]$	$[\text{I}^-]$	$\Delta[\text{S}_2\text{O}_3^{-2}]$
		0.2M	0.2M	0.01M	2%	0.2M	0.2M	(mL)	(s)	(M)	(M)	(M)
		1		2			3	4		5	6	7

1	1	2mL	2mL	2mL	1mL	2mL	2mL	11mL	_____	_____	_____	_____
1	2	2mL	2mL	2mL	1mL	0mL	4mL	11mL	_____	_____	_____	_____
1	3	4mL	0mL	2mL	1mL	2mL	2mL	11mL	_____	_____	_____	_____

2	1	2mL	2mL	2mL	1mL	2mL	2mL	11mL	_____	_____	_____	_____
2	2	2mL	2mL	2mL	1mL	0mL	4mL	11mL	_____	_____	_____	_____
2	3	4mL	0mL	2mL	1mL	2mL	2mL	11mL	_____	_____	_____	_____

8 Average time Δt (s)

- 1 _____
- 2 _____
- 3 _____

Part A. Obtaining the rate law.

Exp.	$\Delta[\text{S}_2\text{O}_3^{-2}]$ (M) (7)	Δt (s) (8)	r (M/s) (9)	$[\text{S}_2\text{O}_8^{-2}]$ (M) (5)	$[\text{I}^-]$ (M) (6)	k (12)
(14) 1	_____	_____	_____	_____	_____	_____
2	_____	_____	_____	_____	_____	_____
3	_____	_____	_____	_____	_____	_____

$$r = k [\text{S}_2\text{O}_8^{2-}]^a [\text{I}^-]^b$$

(10) $a =$ _____(11) $b =$ _____(13) Average k (write down units) = _____

Parts B and C. Evaluating the influence of the temperature on the reaction rate and a catalyst

Exp.	Temperature (K)	$\Delta[\text{S}_2\text{O}_3^{-2}]$ (M) (7)	Δt (s) (8)	r (M/s) (9)	$[\text{S}_2\text{O}_8^{-2}]$ (M) (5)	$[\text{I}^-]$ (M) (6)	k (12)
(14) 1	_____	_____	_____	_____	_____	_____	_____
1	_____	_____	_____	_____	_____	_____	_____

(Results on the first row of Table A and Table B should be the same)

Catalyst 1 _____

(15) $E_a =$ _____

J/K·mol

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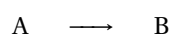
Name:

Date:

Post-lab Questions

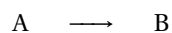
Characterization of the amylase enzyme

1. Use the data below to calculate the rate law of the following reaction:



Experiment	r (M/s)	$[A]$, (M)
1	5×10^{-3}	0.5
2	5×10^{-3}	0.6

2. Use the data below to calculate the rate law of the following reaction:



Experiment	r (M/s)	$[A]$, (M)
1	0.5	0.253
2	1	0.320

3. Use the data below to calculate the rate law:

Experiment	$[A]$ (M)	$[B]$ (M)	r (M/s)
1	0.1	0.02	1.03×10^{-3}
2	0.1	0.04	2.08×10^{-3}
3	0.2	0.01	1.04×10^{-3}
4	0.4	0.01	2.08×10^{-3}

